Gaël BOYA

R&D Engineer — Hardware/Software Co-design (FPGA/SoC/Firmware/RISC-V)

boyaarisgael@gmail.com
Grenoble (38000)
adebayo17.github.io
France
07 53 07 00 18

LANGUAGES

French
Langue maternelle

English
Niveau C1

SOCIAL NETWORKS

Adébayo BOYA
Adébayo BOYA

SKILLS

Hardware Design
Verilog, SystemVerilog, VHDL, RTL Design, Wishbone, AXI4, Vivado, ModelSim, Quartus

Embedded Software
C/C++, Python, Bare-metal drivers, DMA, Cache management

Architectures
RISC-V (RV32I), Zynq PS/PL, AXI DMA, SoC Integration

OS & RTOS
FreeRTOS, Bare-metal, Embedded Linux

Tools
Vivado/Vitis, STM32CubeIDE, PlatformIO, Icarus Verilog, GTKWave, Git, SVN

SOFT SKILLS

End-to-end system thinking

Rigour & methodical debugging

Analytical mindset

Autonomy & full ownership

Clear technical documentation

Dual-degree engineer (Grenoble-INP Phelma & INPT) with a rare dual expertise: digital circuit design (RTL, FPGA/ASIC) and embedded software development (C, RTOS, bare-metal drivers). Capable of owning the full system stack — from RTL specification to firmware validation — with hands-on industrial experience in SoC integration (Silicon Mobility). Seeking an R&D position at the intersection of hardware and software.

EXPERIENCES

Embedded Systems & IoT Trainer (Freelance)

From March 2026 to April 2026 SUPINFO Lyon, France

- Delivered a 35h intensive course: microcontrollers, sensors, communication protocols (UART, I2C) and digital logic circuits.

SoC Integration Intern Engineer

From February 2024 to August 2024 Silicon Mobility Valbonne, France

- RTL design & top-level integration of a reference SoC (FPCU): architecture decisions, IP block integration, exhaustive RTL simulation and functional verification.
- Improved internal SoC integration tooling and authored technical documentation (user guide), streamlining future project delivery timelines.

Engineering Intern in Architecture & Security

From June 2023 to August 2023 Institut National des Postes et Télécommunications Rabat, Morocco

- Microarchitecture modelling of CPU/SoC designs using gem5 simulator; study, simulation and countermeasure implementation against side-channel attacks (Spectre/Meltdown, Flush+Reload).

EDUCATION

Engineering Degree — Integrated Electronic Systems

From September 2023 to April 2025 Grenoble-INP PHELMA Grenoble, France

Engineering Degree — Embedded Systems and Digital Services

From October 2021 to July 2025 Institut National des Postes et Télécommuniations (INPT) Rabat, Morocco

PROJECTS

MiniSoC-RV32I — Full RISC-V Processor

Since August 2025 Personal Grenoble, France

- HW : pipelined 5-stages RV32I core (Verilog) with full hazard unit (forwarding/stall/flush), Wishbone bus, custom peripherals (UART, GPIO, Timer) and complete testbench environment (Icarus Verilog + GTKWave).
- SW : cross-compiled RISC-V GCC toolchain, custom linker script, bare-metal C firmware validated on RTL simulation — complete autonomous HW ↔ SW co-design loop.

AXI DMA Accelerator on Zynq-7020

February 2026 Personal Grenoble, France

- HW: custom AXI4-Stream IP (Verilog, zero-latency bitwise inverter) integrated in Vivado block design: Zynq PS, AXI DMA, AXI Interconnect, IP packaging from scratch.
- SW : bare-metal C driver on ARM Cortex-A9 (Vitis) managing DMA MM2S/SS2M transfers, DDR3 cache flush/invalidation and result verification.

Digital Multi-Effects Audio Processor on FPGA Zynq

From October 2024 to January 2025 Grenoble-INP Phelma Grenoble, France

- HW : Analysed and integrated an existing Vivado Zynq-7020 architecture (Digilent DMA Audio Demo)
- SW : Implemented 5 real-time DSP effects in C on ARM Cortex-A9 (Vitis) : hard/soft clipping distorsion, flanger, delay, chorus; pre-validated all algorithms in Python on real .wav signals.